Ibasho 居場所の向上

Status, explainer and walkthrough

Progress

Phase	State	Notes
1 · Research and direction	DONE	Eight research briefs, plan v2 (VISION, SPEC, ARCHITECTURE, DESIGN_SYSTEM “Andon”, BUILD_PLAN), three red-team reviews, rulings in `docs/review/DECISIONS.md`.
2 · Scaffold and walking skeleton	ACTIVE	G0 monorepo and contracts, then M-0: one thin real slice through every layer.
3 · Parallel build	TODO	Engine, proofs, AI, UI kit, 3D, app shell, assets, landing.
4 · Integration and QA	TODO	Tests, e2e, red-team, polish.

What Ibasho is

Ibasho is a digital twin of a home that you can edit. Change the house and the model recomputes it, using a documented reduced-order heat balance [1], [2]. It searches the retrofit packages you can actually build and shows the best few. Lean 4 [3] proves the bookkeeping. An AI narrator explains results but cannot change a number. The visual language takes cues from architectural storytelling [4].

Features

Feature	State
Planned. Filled in as work packages land.	TODO
Agent link: MCP server and Claude Code plugin, so an AI agent can explore a home headless or live in the app. One-command install.	TODO
G0 scaffold: a 28-package monorepo with enforced layers and ownership. The kernel holds every contract plus exact fixed-point money, exact fractions for the score, SHA-256 hashing [5], a seeded generator [6] and Latin hypercube sampling [7], all tested. Both apps boot in Chromium and Firefox.	DONE

Walkthrough

Written once the app runs end to end.

References

[1] ISO, *ISO 52016-1:2017 Energy performance of buildings. Energy needs for heating and cooling*. 2017.

[2] BRE, ‘The Government's Standard Assessment Procedure for Energy Rating of Dwellings, SAP 10.2’, 2023. [Online]. Available: https://www.gov.uk/guidance/standard-assessment-procedure

[3] L. de Moura and S. Ullrich, ‘The Lean 4 Theorem Prover and Programming Language’, in *CADE 28*, 2021.

[4] D. Lee, ‘Dami Lee, architecture storytelling channel’. [Online]. Available: https://www.youtube.com/@DamiLeeArch

[5] NIST, ‘FIPS PUB 180-4: Secure Hash Standard (SHS)’, 2015. [Online]. Available: https://doi.org/10.6028/NIST.FIPS.180-4

[6] D. Blackman and S. Vigna, ‘Scrambled Linear Pseudorandom Number Generators’, *ACM Transactions on Mathematical Software*, vol. 47, 2021, [Online]. Available: https://doi.org/10.1145/3460772

[7] M. D. McKay, R. J. Beckman, and W. J. Conover, ‘A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output from a Computer Code’, *Technometrics*, vol. 21, 1979, [Online]. Available: https://doi.org/10.2307/1268522